

How to Select a 3D Printer

If you are planning to buy a new 3D printer to speed up product prototyping, here is a guide for choosing the most suitable printer for your application

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The idea of 3D printing technology was first introduced back in 1984. Though 3D printing is now utilised in a variety of applications, this concept is relatively new for the Indian industry.

In the field of electronics

Karan Chaphekar of KCbots says, "3D printing has yet to percolate everyday life, but at present, the main application is in rapid prototyping by an engineering and design team (at a start-up, SME, etc) or by hobbyists who want to make unique things."

A basic 3D printer can be of great advantage for electronic product designers or innovators. According to Paul Anand of Biotz, "The most important part lies in mechanics designs of any electronic gadget they design. In this era, user-centric design is becoming the key differentiator for product optimisation." A 3D printer lets the user create rapid prototypes of different designs, which can be sent for user

feedback and optimisation. It reduces the size of product considerably and saves time and effort.

Attributes to consider

Print speed. The speed of a printer can be defined as the time taken to print a specific distance in the Z-axis. It is usually represented in millimetres per hour or inches per hour. With larger print speed, we get the model printed faster. However, print

It is important to select a model that offers a trade-off between speed and quality of print.

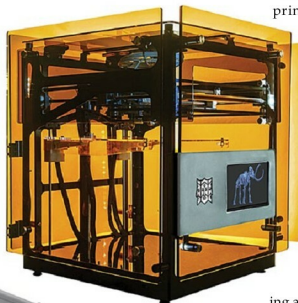
Complexity. Complexity of design that can be printed is determined by three factors: slicer software support, support structure generation and print speed. Some applications require composite design features, such as multiple layers and complex curves. So it is necessary to first define the demands of your specific application.

Print platform. The size of the print platform determines the maximum dimensions that can be printed using a particular printer. It is usually measured in inches or centimetres in XYZ dimensions. For electronics designer, 203×203×203mm would be a good start. Print platforms of 229×229×229mm or more are used in industries. If the printing job can be split into smaller parts and combined later, you can go for smaller build area. Larger print area avoids the need for this scaling and slicing of the print.

Resolution. Resolution can be of two types:

Vertical Z resolution/layer height/layer thickness is usually defined by the thickness of each layer the printer can create in one pass. The smaller the layer thickness, the higher the resolution. Small resolution gives smoother and more realistic print outputs, but this also increases the time to produce the final product as the number of layers increase.

On the other hand, larger resolutions provide models with lesser finish in lesser time. Models that let the user choose layer thickness are also avail-



Brahma3 Anvil 3D printer

speed may depend on some other factors such as complexity of the model to be printed or type of printing material used. Print speed determines how fine and dense your structure will be printed. The speed generally has an inverse relationship with print complexity and quality. The faster the print, the lower the quality. The greater the complexity the lesser the print speed.



CubeX Duo 3D printer